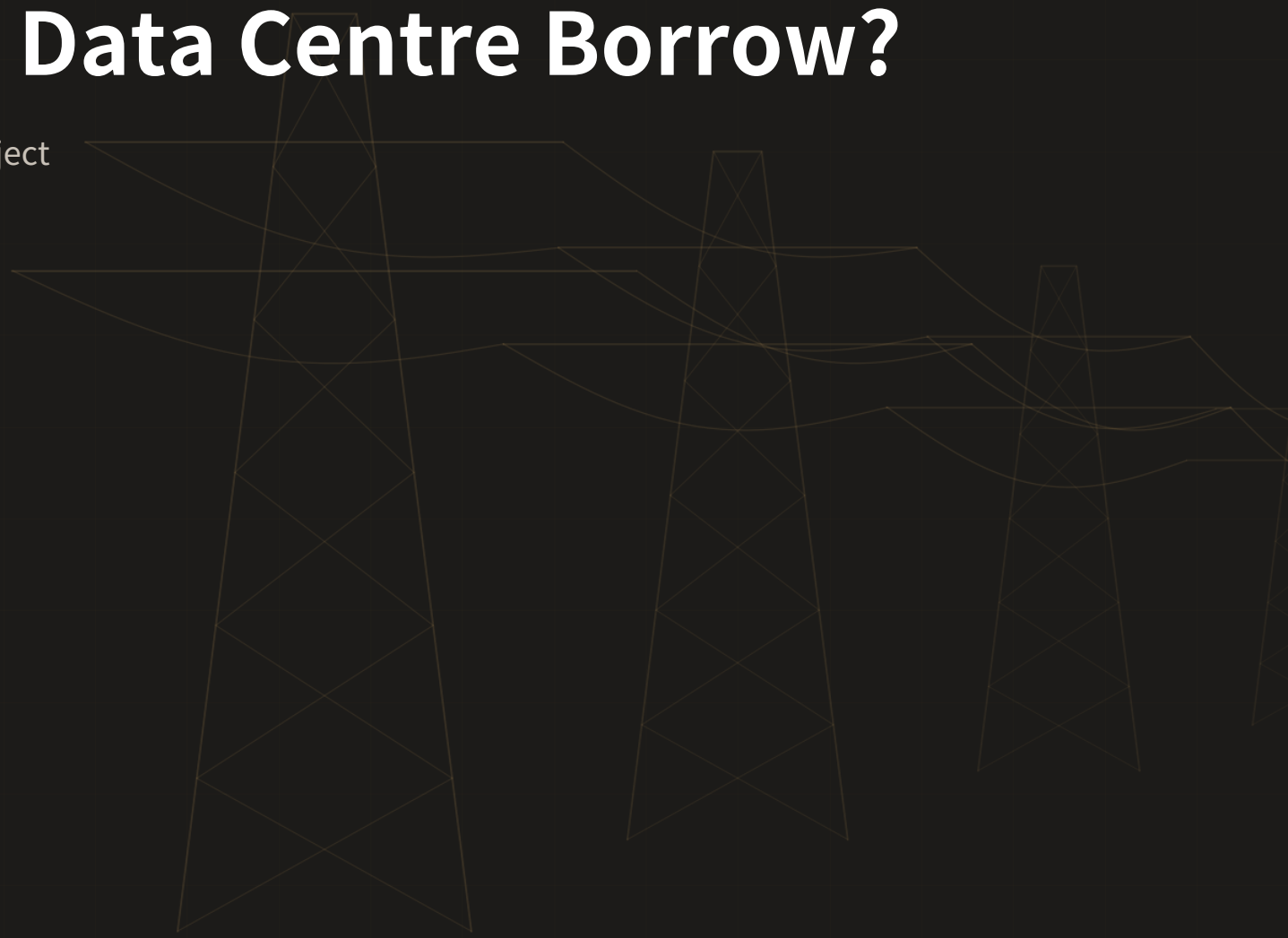
KSJ RESEARCH

Research Note Vol.2

How Much Can an AI Data Centre Borrow?

A lender's view of a 40 MW Korean and a 100 MW U.S. project



Shinjae Kang

2026.10

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Debt capacity: 65% of cost in Korea, 55% in the U.S. — min DSCR is most sensitive to pre-lease ratio (Korea) and rent −20% (U.S.)

One PF engine (sculpting, waterfall, DSRA, LLCR, IRR) run on a 40 MW Korean (KRW) and a 100 MW U.S. (USD) case

Key metrics [*1,*2,*3,*4,*5,*6]

Key metric	Value	p.
Seoul-metro grid approval (2024.8~2026.3)	**1.9% count / 3.0% MW**	1
Seoul-metro pre-leasing (2027–28 deliveries)	16–23.5%	1
Demand growth IEA 2024→2035 (low · base · high)	4.9% · 10.1% · 13.7%	2
KTB 3y · 10y (2026-10-20)	3.84% · 4.18%	2
UST 10y (2026-10-16)	**5.09%**	2
Peer-implied equipment price rise (low · mid · high)	3.3% · 10.9% · 23.9%	2
Korea 40 MW debt (share of cost)	**KRW 134bn (65%)**	3
Korea 40 MW min DSCR · equity IRR	2.16x · 19.0%	3
U.S. 100 MW debt (share of cost)	**USD 1,068mn (55%)**	6
U.S. 100 MW min DSCR · equity IRR	1.30x · 8.3%	6
Phase 2 (20 MW) option value	KRW 27.5bn	5

Source: *1 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026, all new applications; source being verified) *2 Blockmedia (citing CBRE); CBRE Global Data Center Trends 2026; CBRE North America Data Center Trends H1 2026 *3 IEA, Energy and AI (Apr 2025), executive summary *4 Bank of Korea ECOS market rates (daily), USD/KRW *5 FRED, Federal Reserve Bank of St. Louis (DGS10) *6 FSS DART annual reports (consolidated)
Note: KSJ Research. Hypothetical cases; definitions and sources in footnotes and A2–A3. Judgements are the author's own

KRW 134bn

Phase 1 (20 MW) debt (65% of cost)

Korea 40 MW — maximum gearing sets the limit; min DSCR 2.16x

- Min DSCR on contracted rent only: 0.03x (pre-lease 19.8%)
- Seoul-metro grid approval 1.9% by count (10 of 522), 3.0% by capacity (1,010 of 33,592 MW)
- Pre-leasing 16–23.5% for 2027–28 deliveries; rent about KRW 250,000/kW-month

$1,068mn

Debt capacity (KRW 1,422bn; 55% of cost)

U.S. 100 MW — the target DSCR sets the limit; min DSCR 1.30x

- Margin not yet sourced → 250bp scenario shown · margin 150~350bp: USD 1,124mn~USD 1,016mn
- Base-rate gap: UST 10y 5.09% − KTB 3y 3.84% = +1.25pp
- Virginia commercial sales 5.6%/yr (U.S. 0.9%); price 9.55¢/kWh (2025), passed through to the tenant

+10.9%

Three peers' combined implied price rise (FY2022A→FY2025A)

Cost risk — peer margins 5.2%→14.4%, evidence of pricing power

- If all margin gain were price: implied rise 3.3%–23.9% (upper bound, not causal)
- CAPEX KRW 10.0bn → 14.5bn per MW: min DSCR 2.16x → 1.49x; equity IRR 19.0% → 12.5%

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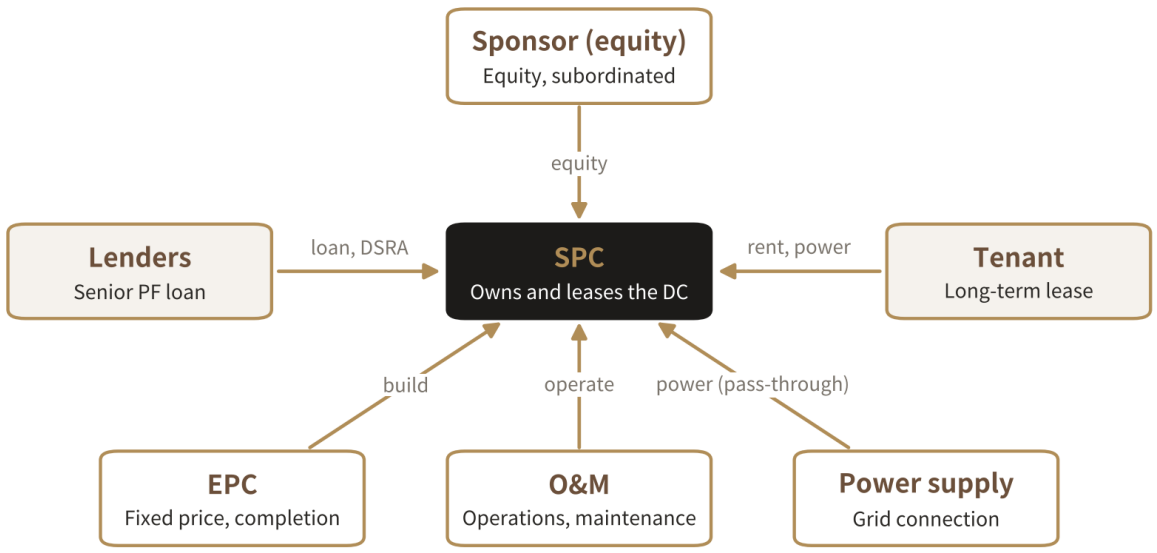
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App. **Appendix** A1–A6
A1 Q&A · A2–A3 assumptions and sources by case · A4 Data notes · A5 Methodology · A6 Glossary

Structure: an SPC with long-term committed leases and power pass-through — Korea 20 MW + 20 MW option, U.S. 100 MW

Case A (main) 40 MW outer Seoul metro, KRW; Case B (comparison) 100 MW Virginia, USD — Seoul-metro grid approval 1.9% by count (10 of 522), 3.0% by capacity (1,010 of 33,592 MW)

The SPC repays debt from long-term committed rent; power is passed through to the tenant *1



Source: *1 KSJ Research
Note: Case A structure (hypothetical). Power pass-through 100%: SPC's net power cost = 0

Five contracts shape the cash flow — lenders watch leasing, rates, cost and grid *1

Contract	Party	Key terms	Lender risk
Lease	Tenant	Long-term committed lease; rent per kW-month; annual escalation	Pre-lease ratio; tenant credit
Loan	Lenders	Target DSCR; max gearing; tenor, grace; DSRA	CFADS volatility; rates
EPC	Contractor	Fixed price; completion date; delay LDs	Equipment prices; delay
Power supply	Utility	Grid-connection approval; capacity	Phase 2 grid approval
Power cost	Tenant	Usage × tariff, 100% pass-through	Pass-through carve-outs

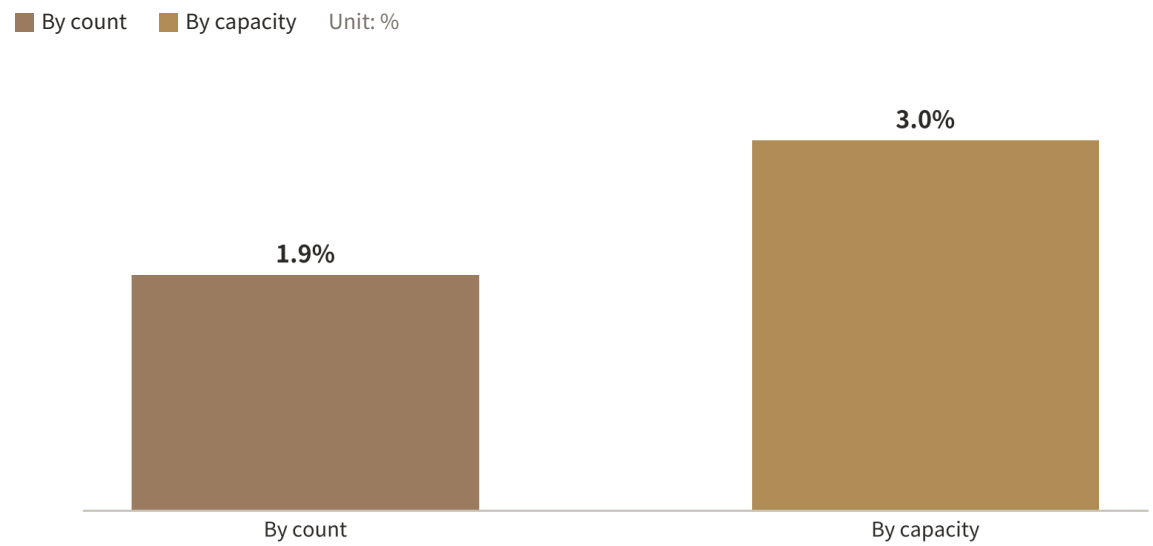
Source: *1 KSJ Research
Note: Hypothetical structure; actual terms vary by project

Korea builds in 20 MW steps, the U.S. in one 100 MW block — same engine, different parameters *1,*2,*3,*4,*5

	Case A (main)	Case B (comparison)
Location	Outer Seoul metro (Incheon, W. Gyeonggi)	Virginia, U.S.
Capacity (IT load)	40 MW = 20 MW firm + 20 MW option	100 MW, single phase
Currency · base rate	KRW · KTB 3y (3.84%)	USD · DGS10 (5.09%)
Rationale	Grid approval 1.9% count / 3.0% MW; grid impact assessment from 10 MW; domestic operating scale ~40 MW	Hyperscale scale; Virginia commercial sales growth (EIA)

Source: *1 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026, all new applications; source being verified) *2 Distributed Energy Act (grid impact assessment) *3 Bank of Korea ECOS market rates (daily), USD/KRW *4 FRED, Federal Reserve Bank of St. Louis (DGS10) *5 U.S. Energy Information Administration (EIA), Electricity retail sales
Note: Not real projects. Base rates: latest business day on or before 2026-10-20

Seoul-metro grid approval is 1.9% count / 3.0% MW — why phase 2 is an option *1

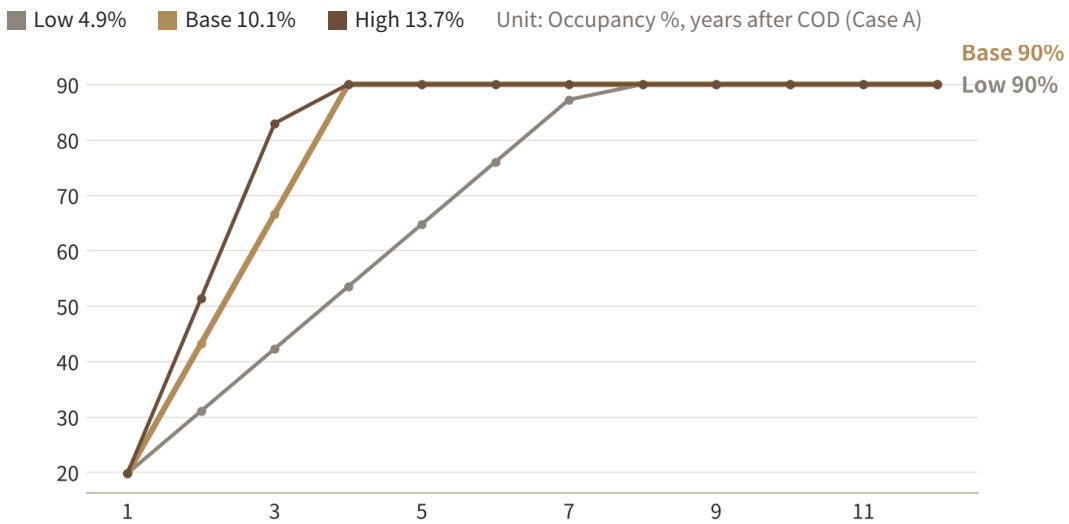


Source: *1 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026, all new applications; source being verified)
Note: Seoul-metro grid approval 1.9% by count (10 of 522), 3.0% by capacity (1,010 of 33,592 MW). 2024.8~2026.3, all new grid applications in the Seoul metro. Projects ≥10 MW need a grid impact assessment

Vol.1 data into PF inputs: base demand growth 10.1%, UST 10y 1.25pp above KTB 3y, peer margins point to pricing power

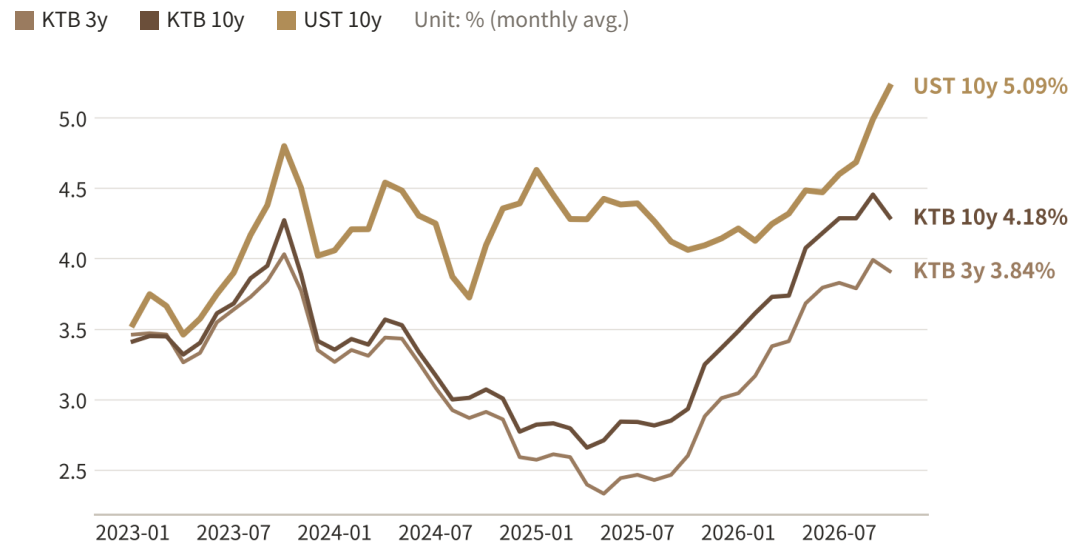
Demand (IEA) → ramp-up · rates (ECOS, FRED) → base rates · peer margins (DART) → CAPEX stress · Virginia (EIA) → Case B demand and power

Pre-lease 19.8% → stabilised 90%: ramp-up 3 yrs (base); IEA growth implies 6.2 (low) and 2.2 yrs (high) *1,*2



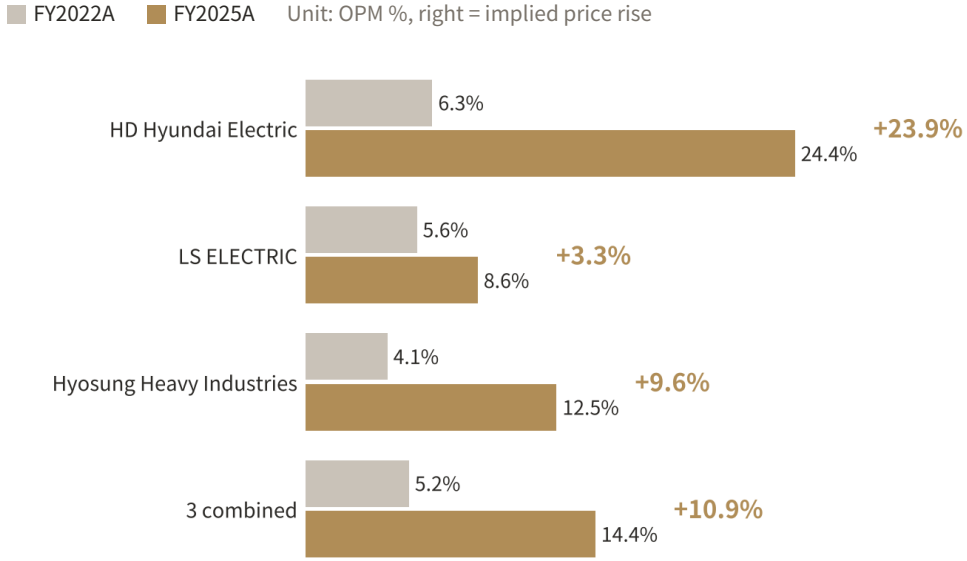
Source: *1 IEA, Energy and AI (Apr 2025), executive summary *2 Blockmedia (citing CBRE); CBRE Global Data Center Trends 2026; CBRE North America Data Center Trends H1 2026
Note: Growth = IEA data-centre power CAGR 2024→2035 (415→low 700, base 1,200, high 1,700 TWh). Scenario ramp-up = base 3 yrs (author's judgment) × base growth ÷ scenario growth

Base rates (2026-10-20): KTB 3y 3.84%, 10y 4.18%; UST 10y 5.09% — model base-rate gap 1.25pp *1,*2



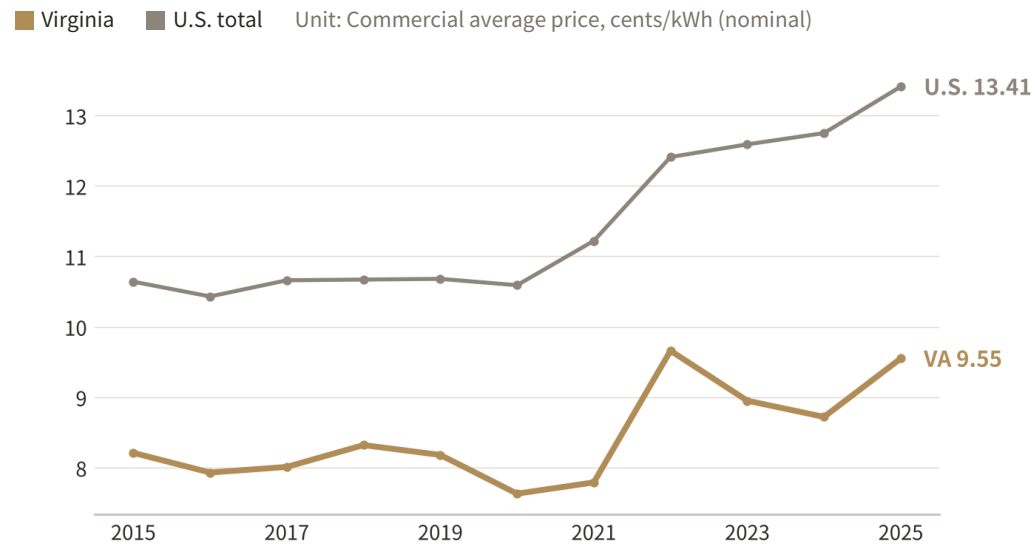
Source: *1 Bank of Korea ECOS market rates (daily), USD/KRW *2 FRED, Federal Reserve Bank of St. Louis (DGS10)
Note: Lines = monthly averages; end labels = as-of values. Model base rates: Korea = KTB 3y; U.S. = UST 10y (2026-10-16). Margins are separate

Peer margins 5.2%→14.4% — if all price, equipment +10.9% (3.3–23.9%) *1



Source: *1 FSS DART annual reports (consolidated)
Note: Implied rise = (1 − start OPM) ÷ (1 − end OPM) − 1: an upper bound if unit costs were flat and all margin gain came from price. Volume, FX and mix are not separated — evidence for a stress, not causation

Virginia: commercial sales 5.6%/yr (U.S. 0.9%); price 9.55¢/kWh (2025) — faster demand, cheaper power than the U.S. *1

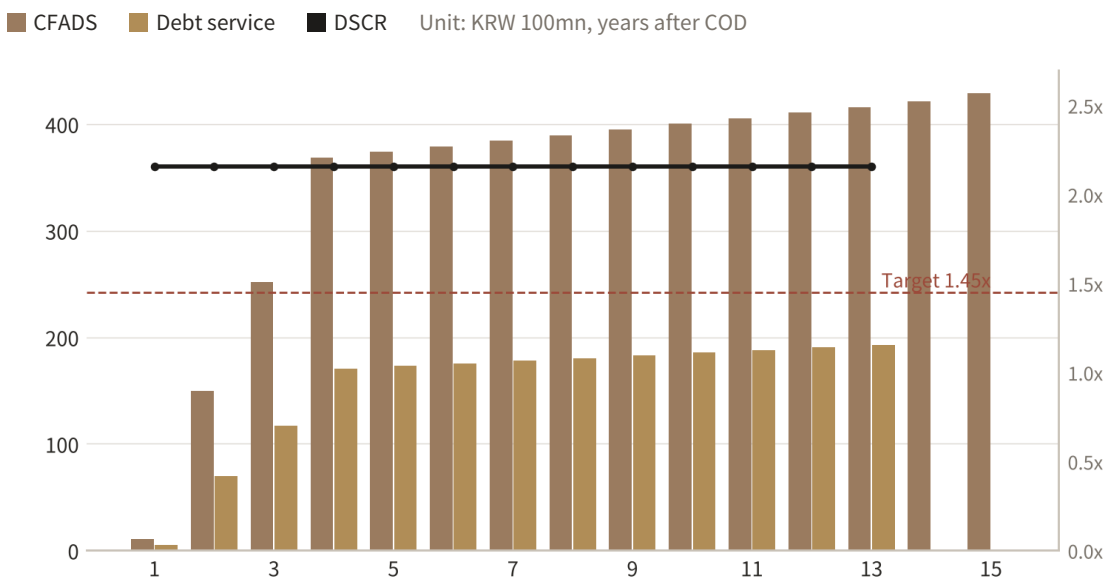


Source: *1 U.S. Energy Information Administration (EIA), Electricity retail sales
Note: Case B demand: Virginia commercial sales 2015→2025 5.6%/yr, +6.2% in 2025. Power price = Virginia 2025. Largest annual rise +24.0% in 2022 (sensitivity on p7)

Korea 40 MW debt capacity KRW 134bn (65% of cost) — maximum gearing sets the limit

Min DSCR 2.16x · LLCR 2.16x · equity IRR 19.0% — on contracted rent only, min DSCR 0.03x

Sculpting debt service at CFADS ÷ 1.45x gives KRW 200bn [*1]



Source: *1 KSJ Research
Note: CFADS = EBITDA − tax. Tenor = 2 yrs construction (grace) + 13 yrs repayment after COD; extra grace after COD 0 yrs. Base scenario. No depreciation (life not set → tax conservative)

Debt KRW 134bn — maximum gearing sets the limit [*1]

Item	Value
Sculpting limit (target DSCR)	KRW 200bn
Gearing limit (65%)	KRW 134bn
Debt adopted (lower)	**KRW 134bn**
Binding constraint · effective DSCR	maximum gearing sets the limit · 2.16x
Total cost = CAPEX + IDC + initial DSRA	KRW 206bn = KRW 200bn + KRW 6.2bn + KRW 0.2bn
Equity · gearing	KRW 72.2bn · 65.0%
Min · avg DSCR · LLCR	2.16x · 2.16x · 2.16x
Project IRR · equity IRR · payback (from FC)	13.0% · 19.0% · 7.7 yrs
Check: debt if interest is paid in cash every year	KRW 13.6bn

Source: *1 KSJ Research
Note: Debt = min(sculpting, max gearing × total cost); cost includes IDC and initial DSRA, solved iteratively. In year(s) 1, 2 debt service is below interest (negative principal); without a ramp-up interest reserve the last row is the realistic limit

Waterfall: CFADS → interest → principal → DSRA (6 months) → distributions [*1]

Unit: KRW 100mn

Year	CFADS	Interest	Principal	DSRA net	Distrib.	DSCR
1	10	72	-67	6	0	2.16x
2	150	75	-6	50	30	2.16x
3	252	76	41	27	108	2.16x
5	374	68	105	1	199	2.16x
10	400	36	150	1	213	2.16x
13	416	10	183	-96	320	2.16x

Source: *1 KSJ Research
Note: DSRA target = next year's debt service × months/12, released at maturity. Distributions are locked while interest is capitalised and released after grace. Negative principal = debt service below interest

Debt barely moves across demand scenarios (KRW 134bn, gearing cap); min DSCR 1.83x–2.23x [*1,*2]

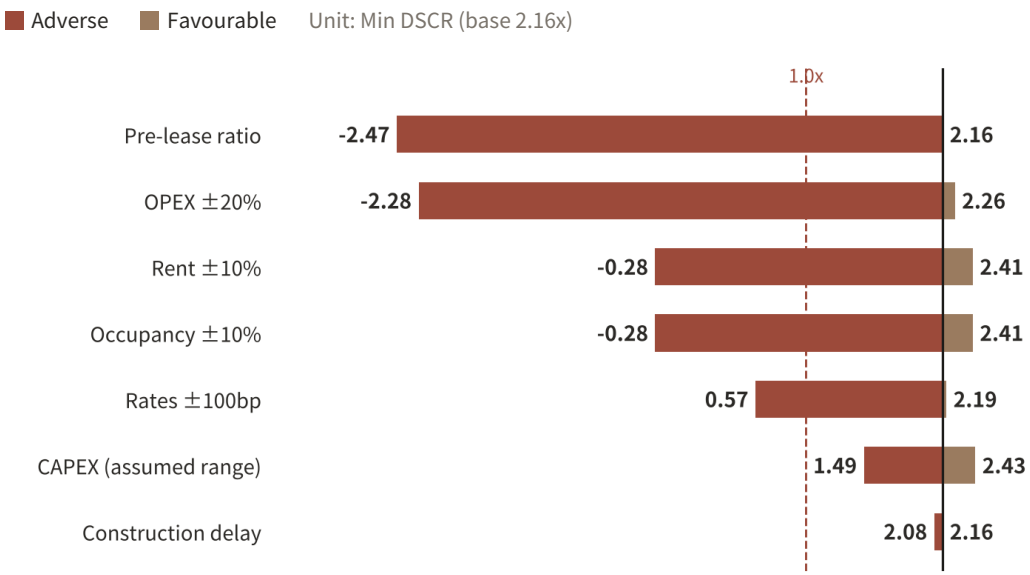
Scenario (growth)	Debt	Min DSCR	LLCR	Project IRR	Equity IRR	Contracted DSCR
Low (ramp 6.2 yrs)	KRW 134bn	1.83x	1.83x	11.4%	16.0%	0.02x
Base (ramp 3.0 yrs)	KRW 134bn	2.16x	2.16x	13.0%	19.0%	0.03x
High (ramp 2.2 yrs)	KRW 134bn	2.23x	2.23x	13.4%	19.8%	0.03x

Source: *1 IEA, Energy and AI (Apr 2025), executive summary *2 KSJ Research
Note: Debt re-sized per scenario. Ramp-up = base period × IEA base growth ÷ scenario growth. Contracted DSCR = min DSCR on pre-leased rent only

Stress: pre-lease ratio moves min DSCR the most — base 2.16x

Case A in full, Case B summarised · breakevens at min DSCR 1.0x and target · tornado · heatmap (rent × CAPEX)

Pre-lease ratio moves min DSCR the most (-2.47x–2.16x) *1,*2



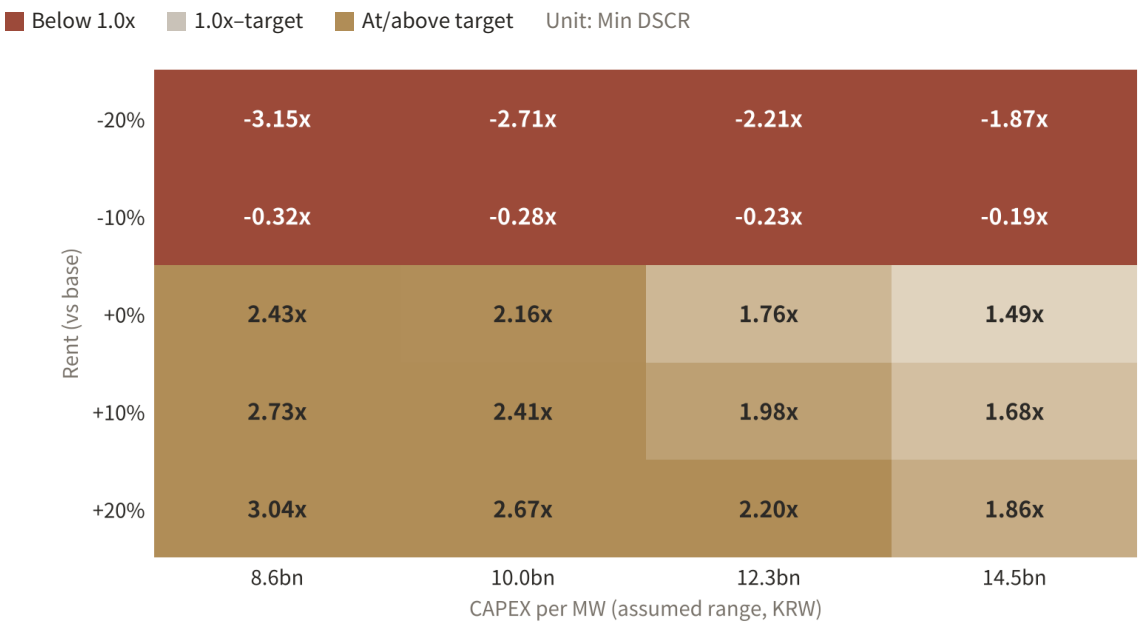
Source: *1 KSJ Research *2 FSS DART annual reports (consolidated)
Note: Debt schedule fixed at base. Delay = +4 quarters; pre-lease 16–23.5%; CAPEX = KRW 8.6–14.5bn per MW; OPEX ±20% (unsourced judgment). Negative DSCR = negative CFADS.

Breakevens where min DSCR falls below target (1.45x) and 1.0x *1

Variable	Base	Target 1.45x	1.0x
Rent (vs base)	0%	-3%	-5%
Occupancy (vs base)	0%	-3%	-5%
Pre-lease ratio	19.8%	19.2%	18.8%
CAPEX (vs base)	0%	+49%	+116%
OPEX (vs base)	0%	+3%	+5%
Base-rate change	0bp	+18bp	+42bp
Construction delay	0 qtrs	none in range	none in range

Source: *1 KSJ Research
Note: One variable at a time; debt schedule fixed (extra CAPEX and delay interest funded at base gearing); nearest crossing to base. 'Base = target': sculpting already sets base min DSCR at target. 'None in range': no crossing in the search range

Rent -20% with CAPEX KRW 14.5bn/MW: min DSCR -1.87x *1



Source: *1 KSJ Research
Note: Extra CAPEX funded at base gearing (debt and principal scaled)

Case B: min DSCR 1.30x; most sensitive: Rent −20%; rent −20%: 1.00x *1

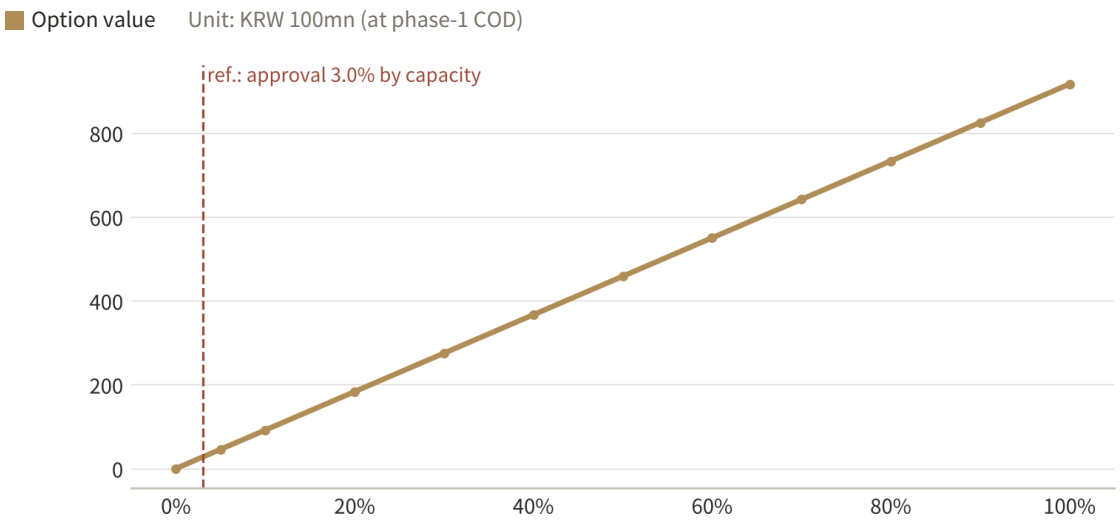
Variable	Adverse	Favourable
Rent −20%	1.00x	1.30x
Rent ±10%	1.15x	1.43x
Occupancy ±10%	1.15x	1.37x
Rates ±100bp	1.20x	1.32x
OPEX ±20%	1.25x	1.34x
Construction delay	1.25x	1.30x

Source: *1 KSJ Research
Note: Same stresses plus rent −20% (100 MW hyperscale leases may price below 250–500 kW quotes). Debt schedule fixed; 250bp margin scenario (not yet sourced)

Phase 2 (20 MW): option value KRW 27.5bn at p = 30% — the right to wait adds KRW 2.4bn

Approval 10/30/50% · decision at COD+3 yrs · ref.: Seoul-metro approval 1.9% count / 3.0% MW · discount rate and window on a grid

Option value at approval probability 10%/30%/50%: KRW 9.2bn / KRW 27.5bn / KRW 45.8bn [*1,*2]



Source: *1 KSJ Research *2 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026, all new applications; source being verified)
Note: Approval scenarios 10/30/50% (base 30%, author's judgment). The approval rate covers all new applications, not one project's odds. Value = p × conditional value. discount rate and waiting window not yet sourced → middle grid cell (10%, 2 yrs)

Build now KRW 25.1bn vs wait KRW 27.5bn — value of waiting KRW 2.4bn [*1]

Item (p = 30%)	Value
Build now (on approval)	KRW 25.1bn
Wait (optimal timing, option value)	**KRW 27.5bn**
Flexibility = option − max(now, 0)	**KRW 2.4bn**
Trigger rent (Low)	no build at decision date
Trigger rent (Base)	KRW 254,524/kW-mo
Trigger rent (High)	KRW 254,524/kW-mo
NPV-zero rent (base)	KRW 178,974/kW-mo
Share of paths built · mean build year	85% · 3.9 yrs

Source: *1 KSJ Research
Note: Decision year 3 after phase-1 COD (= end of ramp-up, when demand is confirmed); window 2 yrs; discount 10%; rent vol. 15% (judgment); 20,000 paths, seed 42: s.e. 0.6% (1,000 paths: 2.6%) — converged

Option value KRW 15.3–46.0bn across discount rates and windows — waiting adds flexibility [*1]

Unit: KRW bn; brackets = flexibility (p = 30%)

Discount \ window	0 yrs	2 yrs	4 yrs
8%	43.7 (0.3)	45.1 (1.9)	46.0 (2.7)
10%	26.5 (0.8)	**28.0 (2.5)**	29.0 (3.4)
12%	15.3 (1.8)	17.0 (3.6)	17.9 (4.5)

Source: *1 KSJ Research
Note: Discount rate and window are not yet sourced, so a sensitivity grid is shown (author-set axes). Gold = headline cell. 8,000 paths per cell

Decision structure: grid approval (p) → observe rent and demand → build now or wait [*1]

Phase 2 (20 MW) = the right to wait for grid approval and demand

- Uncertainty ①: grid approval with probability p at the decision date; zero value if refused
- Uncertainty ②: rent follows geometric Brownian motion (drift = escalation, volatility σ)
- Uncertainty ③: IEA low/base/high ramp-up drawn per path — observed via phase 1 occupancy
- Phase 2 pre-lease = phase 1 occupancy in the decision year (build after demand is confirmed)
- Valuation: Monte Carlo + LSM (least-squares continuation value), fixed seed, convergence by path count

Source: *1 KSJ Research
Note: Unlevered NPV. Option value = p × value given approval

Korea vs U.S.: different bottlenecks set debt capacity — grid and pre-leasing in Korea, scale and rates in the U.S.

Base-rate gap +1.25pp (UST 10y − KTB 3y) · grid approval 1.9% count / 3.0% MW · pre-leasing 16–23.5% · FX 1,332.1

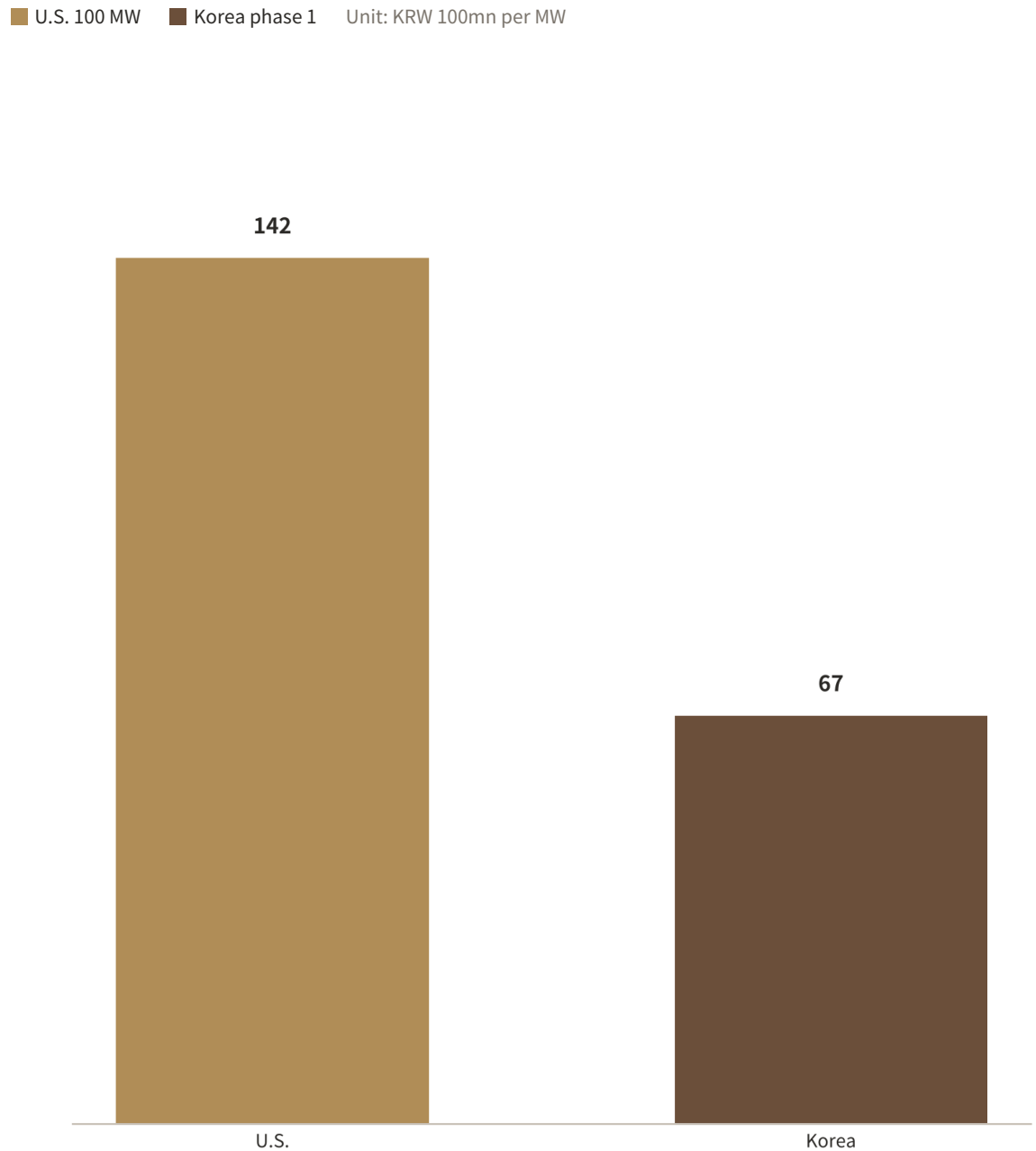
Debt, DSCR and equity IRR side by side — same engine, different bottlenecks

*1,*2,*3,*4,*5,*6,*7

	Korea 40 MW (Case A)	U.S. 100 MW (Case B)
Capacity · phasing	40 MW (20 firm + 20 option)	100 MW single
Currency · base rate	KRW · KTB 3y 3.84%	USD · UST 10y 5.09%
Margin	150bp	Not sourced: 250bp scenario (150/250/350bp)
Debt (local currency)	KRW 134bn	USD 1,068mn (USD 1,016mn–USD 1,124mn)
Debt (KRW)	KRW 134bn	KRW 1,422bn
Debt per MW (KRW)	KRW 6.7bn	KRW 14.2bn
Gearing · binding	65% · GEARING	55% · DSCR
Min · avg DSCR	2.16x · 2.16x	1.30x · 1.30x
LLCR	2.16x	1.30x
Project IRR	13.0%	7.7% (7.7%–7.7%)
Equity IRR	19.0%	8.3% (7.8%–8.7%)
Key bottleneck	**Grid (approval 1.9% count / 3.0% MW) · pre-leasing (16–23.5%)**	**Scale (100 MW, 5x Korea phase 1) · rates (+1.25pp)**

Source: *1 Bank of Korea ECOS market rates (daily), USD/KRW *2 FRED, Federal Reserve Bank of St. Louis (DGS10) *3 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026, all new applications; source being verified) *4 Blockmedia (citing CBRE); CBRE Global Data Center Trends 2026; CBRE North America Data Center Trends H1 2026 *5 Etoday (citing Daol Investment & Securities, 1 Oct 2026); Edaily (12 Oct 2026); Cushman & Wakefield 2026 Data Center Development Cost Guide *6 Gridlines (2026); Lev (20 Aug 2026); Edaily (12 Oct 2026) *7 KSJ Research
Note: KRW = USD mn × ECOS USD/KRW 1,332.1 (2026-10-20) ÷ 100. Korea debt is for phase 1 (20 MW). Case B brackets = 150–350bp margin range (margin not yet sourced). No depreciation → IRRs conservative

Debt per MW: U.S. KRW 14.2bn vs Korea KRW 6.7bn *1,*2



Source: *1 Bank of Korea ECOS market rates (daily), USD/KRW *2 KSJ Research
Note: USD debt converted at the as-of rate (no hedging cost). CAPEX per MW (KRW 10bn vs USD 18.2mn) explains most of the gap

Implication: for Korean AI data-centre PF, 'when' comes before 'how much' — grid approval and an anchor tenant first

Lender checklist · limitations · next: Vol.3 (IPO pricing, overseas peers) · Case B power-cost sensitivity (Virginia prices)

Lender checklist: Korea — approval and tenants first; U.S. — rates and cost first [*1,*2,*3,*4,*5]

Item	Korea 40 MW	U.S. 100 MW	Pages
Grid approval	Phase 1 approval a must; phase 2 as an option (1.9% count / 3.0% MW)	Confirm connection timing	1, 5
Pre-leasing	Breakeven for 1.0x: 18.8% (market 16–23.5%)	Pre-lease 80%; 15-yr lease	3, 4
Rates	KTB 3y 3.84% + 150bp	UST 10y 5.09% + margin TBC	2, 6
Construction cost	KRW 8.6–14.5bn per MW	Large absolute exposure	2, 4
OPEX	Unsourced judgment → ±20% stress	Unsourced judgment → ±20% stress	4
Power cost	Pass-through clause; carve-outs	Tenant burden as Virginia prices rise (table below right)	2, 7

Source: *1 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026, all new applications; source being verified) *2 Blockmedia (citing CBRE); CBRE Global Data Center Trends 2026; CBRE North America Data Center Trends H1 2026 *3 Etoday (citing Daol Investment & Securities, 1 Oct 2026); Edaily (12 Oct 2026); Cushman & Wakefield 2026 Data Center Development Cost Guide *4 Gridlines (2026); Lev (20 Aug 2026); Edaily (12 Oct 2026) *5 KSJ Research

Note: Checklist for the hypothetical cases; figures from the body

Limitations of this note

- Both cases are hypothetical; author's judgments are flagged as such in A2–A3
- Case A early ramp-up (year 1, 2): debt service < interest; without an interest reserve the limit is KRW 13.6bn
- Depreciation life is not set, so no depreciation is taken → tax overstated, IRRs conservative
- No terminal value → equity IRR uses 25 years of cash flow only
- PUE and IT load not set → power is a 100% pass-through; sensitivity is shown as tenant burden
- Peer margin → price is an upper bound, not causal (volume, FX, mix not separated)

Note: Updated as sources are confirmed

Next in Vol.3: IPO pricing and overseas peers

A hypothetical unlisted power-equipment IPO in a re-rated sector: what price range?

- Korean peers' P/E and EV/EBITDA + overseas peers (SEC EDGAR)
- Market cap (listed parent) vs financials (filer) checked for all five
- Fair value → IPO discount → indicative price range · Monte Carlo sensitivity

Note: Timing and method of Vol.3 may change

At Virginia's 9.55¢, power costs 44–68% of rent — pass-through shields the SPC, not the tenant [*1,*2]

Unit: USD per IT kW-month (brackets = share of USD 190 rent)

PUE	Price 2025	Price 2025 +24.0%
1.2	$84 (44%)	$104 (55%)
1.3	$91 (48%)	$112 (59%)
1.4	$98 (51%)	$121 (64%)
1.5	$105 (55%)	$130 (68%)

Source: *1 U.S. Energy Information Administration (EIA), Electricity retail sales *2 Blockmedia (citing CBRE); CBRE Global Data Center Trends 2026; CBRE North America Data Center Trends H1 2026
Note: Power = 730 h × PUE × price (100% IT load). Price = EIA Virginia commercial 2025; right column applies the largest annual rise (2022). PUE is a sensitivity axis, not an assumption

Expected Q&A: what lenders ask first

Questions likely in credit review and lender due diligence, with page references

8 expected questions, including the lender's view *1,*2,*3,*4,*5,*6,*7

Question	Answer	Page
Can it borrow without an anchor tenant?	On contracted rent only, min DSCR is 0.03x (pre-lease 19.8%); the 1.0x breakeven pre-lease is 18.8%. Lenders need contracted cash flow to cover debt service, so without an anchor tenant capacity shrinks sharply or disappears	2, 3, 4
Why 40 MW in Korea?	Seoul-metro grid approval 1.9% by count (10 of 522), 3.0% by capacity (1,010 of 33,592 MW). Projects of 10 MW or more need a grid impact assessment, and domestic operating sites are around 40 MW. Hence 20 MW firm plus a 20 MW option built after approval and demand are confirmed	1, 5
Why 100 MW in the U.S., and what differs?	Virginia commercial sales grow 5.6%/yr vs U.S. 0.9%. But its base rate (UST 10y) is 1.25pp above KTB 3y, so debt service is heavier at the same DSCR	2, 6
What if rates rise 100bp?	With the schedule fixed, min DSCR 2.16x → 0.57x; equity IRR 19.0% → 18.3%	4
What if equipment prices rise?	If the three peers' margin rise 5.2%→14.4% (FY2022A→FY2025A) were all price, equipment prices rose 3.3%–23.9% (upper bound, not causal).	2, 4
Who pays for power?	A 100% pass-through: the tenant pays usage × tariff, so the SPC's net power cost is zero and tariff risk becomes a tenant-affordability question (Virginia commercial average 9.55¢/kWh, 2025; sensitivity on p7)	1, 2
Build phase 2 now or wait?	At p = 30%: build now KRW 25.1bn vs wait KRW 27.5bn; the right to wait is worth KRW 2.4bn	5
How is FX handled for the U.S. case?	Computed in USD, converted at ECOS USD/KRW 1,332.1 (2026-10-20): KRW 100mn = USD mn × FX ÷ 100. Hedging costs are excluded	6, A4

Source: *1 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026, all new applications; source being verified) *2 Blockmedia (citing CBRE); CBRE Global Data Center Trends 2026; CBRE North America Data Center Trends H1 2026 *3 IEA, Energy and AI (Apr 2025), executive summary *4 Bank of Korea ECOS market rates (daily), USD/KRW *5 FRED, Federal Reserve Bank of St. Louis (DGS10) *6 FSS DART annual reports (consolidated) *7 U.S. Energy Information Administration (EIA), Electricity retail sales
Note: Numbers match the body; shaded = core lender questions

Assumptions and sources: Korea 40 MW — sourced vs judgment

Case A — 40 MW leased data centre SPC, outer Seoul metro (Incheon / western Gyeonggi)

Korea 40 MW: 5 sourced · 16 author's judgment · 6 not set — judgments shaded [*1,*2,*3,*4,*5]

Item	Value (range)	Unit	Basis / source	Status
Phase 1 capacity	20	MW	Hypothetical case design	Design
Phase 2 (option) capacity	20	MW	Hypothetical case design	Design
CAPEX per MW	100 (86–145)	KRW 100mn/MW	Author's judgment; ref. Etoday (citing Daol Investment & Securities), Edaily	**Judgment**
Construction period	8	quarters	Author's judgment	**Judgment**
Rent (per kW-month)	250,000	KRW/kW-month	Blockmedia (citing CBRE)	**Sourced**
Rent escalation	2% (2–3%)	%/yr	Blockmedia (citing CBRE)	**Sourced**
Pre-lease ratio	19.75% (16–23.5%)	%	Blockmedia (citing CBRE)	**Sourced**
Stabilised occupancy	90%	%	Author's judgment	**Judgment**
OPEX (% of stabilised revenue, ex power)	20%	%	Author's judgment	**Judgment**
OPEX escalation	2.5%	%/yr	Author's judgment	**Judgment**
Ramp-up period	3	yrs	Author's judgment	**Judgment**
Tenor basis	fc	-	Hypothetical case design	Design
Corporate tax brackets (national)	10/20/22/25%	KRW 100mn:%	Taxguide; Corporate Tax Act Art. 55	**Sourced**
PUE	—	x	Not set → 100% pass-through, SPC net power cost zero	Input needed
IT load factor	—	%	Not set → 100% pass-through, SPC net power cost zero	Input needed
Power pass-through	100%	%	Hypothetical case design	Design
Local/state tax	10%	% of national tax	Taxguide; Corporate Tax Act Art. 55	**Sourced**
Depreciation life	—	yrs	Not set → no depreciation (tax conservative)	Input needed
Operating period	25	yrs	Author's judgment	**Judgment**
Exit cap rate	—	%	Not set → no terminal value	Input needed
Credit margin	150	bp	Author's judgment; ref. Edaily	**Judgment**
Target DSCR	1.45	x	Author's judgment; ref. Gridlines 2026	**Judgment**
Maximum gearing	65%	%	Author's judgment; ref. Edaily	**Judgment**
Loan tenor	15	yrs	Author's judgment	**Judgment**
Grace period	0	yrs	Author's judgment	**Judgment**
Grace-period interest	paid	-	Hypothetical case design	Design
DSRA months	6	months	Author's judgment	**Judgment**
Discount rate (equity)	—	%	Not sourced → sensitivity grid 8%/10%/12%	Input needed
Grid-approval probability	30% (10–50%)	%	Author's judgment	**Judgment**
Approval decision year	3	yrs after phase-1 COD	Author's judgment	**Judgment**
Option window	—	yrs	Not sourced → sensitivity grid 0/2/4 yrs	Input needed
Rent volatility	15%	%/yr	Author's judgment	**Judgment**

Source: *1 Blockmedia (citing CBRE); CBRE Global Data Center Trends 2026; CBRE North America Data Center Trends H1 2026 *2 Etoday (citing Daol Investment & Securities, 1 Oct 2026); Edaily (12 Oct 2026); Cushman & Wakefield 2026 Data Center Development Cost Guide *3 Gridlines (2026); Lev (20 Aug 2026); Edaily (12 Oct 2026) *4 Korean Corporate Tax Act Art. 55 (Taxguide); 26 U.S.C. § 11(b); Va. Code § 58.1-400 *5 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026, all new applications; source being verified)
Note: Status: sourced (gold) = public source · judgment (shaded, brown) = author's setting informed by sources, stress-tested · input needed (red) = not set, excluded or treated conservatively · verify = public basis, not yet checked against the original · design = hypothetical case setting · data = official statistics

Assumptions and sources: U.S. 100 MW — sourced vs judgment

Case B — 100 MW hyperscale leased data centre, Virginia, U.S.

U.S. 100 MW: 7 sourced · 9 author's judgment · 7 not set — judgments shaded [*1,*2,*3,*4]

Item	Value (range)	Unit	Basis / source	Status
Phase 1 capacity	100	MW	Hypothetical case design	Design
CAPEX per MW	18.18	USD mn/MW	Cushman & Wakefield 2026 Cost Guide · 2026-09-03	Sourced
Power equipment share of CAPEX	—	%	Being sourced	Input needed
Construction period	8	quarters	Author's judgment	Judgment
Rent (per kW-month)	190 (190–235)	USD/kW-month	CBRE Global DC Trends 2026 · 2026-06-17	Sourced
Rent escalation	3%	%/yr	Blockmedia (citing CBRE), Gridlines 2026	Sourced
Pre-lease ratio	80%	%	CBRE NA DC Trends H1 2026	Sourced
Stabilised occupancy	95%	%	Author's judgment; ref. CBRE NA DC Trends H1 2026	Judgment
OPEX (% of stabilised revenue, ex power)	15%	%	Author's judgment	Judgment
OPEX escalation	3%	%/yr	Author's judgment	Judgment
Ramp-up period	1	yrs	Author's judgment	Judgment
Lease term	15	yrs	Blockmedia (citing CBRE), Gridlines 2026	Sourced
Tenor basis	fc	-	Hypothetical case design	Design
PUE	—	x	Not set → 100% pass-through, SPC net power cost zero	Input needed
IT load factor	—	%	Not set → 100% pass-through, SPC net power cost zero	Input needed
Power price	0.0955	USD/kWh	EIA commercial average price, Virginia, 2025	Data
Power pass-through	100%	%	Hypothetical case design	Design
Corporate tax (national/federal)	21%	%	26 U.S.C. § 11(b)	Verify
Local/state tax	6%	%	Va. Code § 58.1-400	Verify
Depreciation life	—	yrs	Not set → no depreciation (tax conservative)	Input needed
Operating period	25	yrs	Author's judgment	Judgment
Exit cap rate	—	%	Not set → no terminal value	Input needed
Credit margin	—	bp	Not sourced → 150/250/350bp scenarios	Input needed
Target DSCR	1.3	x	Gridlines 2026	Sourced
Maximum gearing	75%	%	Lev 2026 · 2026-08-20	Sourced
Loan tenor	15	yrs	Author's judgment	Judgment
Grace period	0	yrs	Author's judgment	Judgment
Grace-period interest	paid	-	Hypothetical case design	Design
DSRA months	6	months	Author's judgment	Judgment
Discount rate (equity)	—	%	Being sourced	Input needed

Source: *1 Blockmedia (citing CBRE); CBRE Global Data Center Trends 2026; CBRE North America Data Center Trends H1 2026 *2 Etoday (citing Daol Investment & Securities, 1 Oct 2026); Edaily (12 Oct 2026); Cushman & Wakefield 2026 Data Center Development Cost Guide *3 Gridlines (2026); Lev (20 Aug 2026); Edaily (12 Oct 2026) *4 Korean Corporate Tax Act Art. 55 (Taxguide); 26 U.S.C. § 11(b); Va. Code § 58.1-400
Note: Status: sourced (gold) = public source · judgment (shaded, brown) = author's setting informed by sources, stress-tested · input needed (red) = not set, excluded or treated conservatively · verify = public basis, not yet checked against the original · design = hypothetical case setting · data = official statistics

Data notes: sources, datasets, dates and model inputs

The same official-source data as Vol.1, linked to PF inputs · overseas peers (SEC) in Vol.3

Data sources · datasets · as-of dates

Source	Dataset	Items	As of	Status
IEA	Energy and AI (Apr 2025), Key Questions on Energy and AI (Apr 2026)	Data-centre power, 2024 and 2035 scenarios	2026-10-21	Collected
ECOS · Bank of Korea	Market rates (daily), KRW exchange rates	KTB 3y/10y, USD/KRW	2026-10-20	Collected
FRED · St. Louis Fed	10-Year Treasury Constant Maturity (DGS10)	UST 10y	2026-10-20	Collected
DART · FSS	Annual reports, consolidated	Three peers' revenue and operating income	2026-10-20	Collected
EIA · U.S. EIA	Electricity retail sales (state/sector, annual)	Virginia and U.S. commercial sales and price	2026-10-20	Collected
EIA · U.S. EIA	Electricity retail sales (Virginia commercial price)	Virginia commercial average price	2026-10-20	Collected

Note: As-of = collection date (IEA: date checked against source). Overseas peers (SEC) are used in Vol.3

Data values used in the model, with dates

Item	Value	Date / period	Source
KTB 3y · 10y	3.840% · 4.176%	2026-10-20	ECOS
UST 10y (DGS10)	5.09%	2026-10-16	FRED
USD/KRW	1,332.1	2026-10-20	ECOS
Demand growth (low·base·high)	4.87% · 10.13% · 13.68%	2024→2035	IEA
3-peer combined OPM	5.16% → 14.44%	FY2022A→FY2025A	DART
Grid approval (count)	10 ÷ 522 = 1.92%	2024.8~2026.3	Public reports (verifying)
Grid approval (capacity)	1,010 ÷ 33,592 = 3.01%	2024.8~2026.3	Public reports (verifying)
Virginia commercial average price	9.55¢/kWh	2025	EIA

Note: Latest business day on or before 2026-10-20. KRW = USD mn × USD/KRW ÷ 100

Methodology: one engine, case parameters, verified calculations

Every figure is calculated from data and assumptions, and conclusions are generated from the results

Data links → PF model → stress and option → checks and report [*1]

1. Data links

IEA → ramp-up

ECOS · FRED → base rates

DART → CAPEX stress

EIA → Case B

2. PF model

Quarterly build, IDC

CFADS → sculpting

Gearing cap

Waterfall · DSRA

3. Stress, option

Breakevens, tornado

Heatmap

LSM Monte Carlo

Fixed seed, convergence

4. Checks, report

Accounting identities

DSCR, IRR, FX checks

Conclusion titles

Korean & English

Built with an AI coding tool (Claude Code) for data links, the PF model, Monte Carlo and report generation; case design, assumption sourcing and number checks by the author.

Source: *1 KSJ Research
Note: Full code and data sources: GitHub https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio

Glossary: the lender's vocabulary

Units: KRW 100mn, USD mn · %, x

Glossary

Term	Definition
CFADS	Cash flow available for debt service = EBITDA − tax
DSCR	Debt service cover ratio = CFADS ÷ (interest + principal)
LLCR	Loan life cover ratio = PV of CFADS over loan life ÷ debt
Sculpting	Sizing debt by setting each year's debt service = CFADS ÷ target DSCR
Maximum gearing	Cap on debt ÷ total project cost (LTC)
DSRA	Debt service reserve account: months of next year's debt service
IDC	Interest during construction, capitalised into project cost
Grace period	Principal holiday (interest paid or capitalised)
Committed lease	Long-term lease paying for contracted capacity regardless of use
Pass-through	Clause passing power cost to the tenant at usage × tariff
MW (IT)	Capacity for IT equipment; total facility = IT × PUE
PUE	Total facility power ÷ IT power
Project IRR / Equity IRR	IRR of unlevered after-tax cash flow / of equity cash flow
Real option	Value of the right to defer an investment
LSM	Least-squares Monte Carlo: regression estimates continuation value to time exercise
Grid impact assessment	Pre-assessment of grid impact for large loads (10 MW+)

Note: Shaded = core lender metrics

Disclosures

- This is a personal portfolio for education and research. It is not investment or financial advice.
- Both cases are hypothetical, not real projects; the same model engine runs with different assumptions.
- Views are the author's own and do not represent any current or future employer.
- All data are public; accuracy and completeness are not guaranteed. Assumptions and their sources are in Appendix A2–A3.
- As of 2026-10-20. Methods and code can be reproduced from the repository.

KSJ Research

https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio